Differential-Drive Robot with the CodeCell C6 Drive

Design, Control & Wireless Operation

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Outline

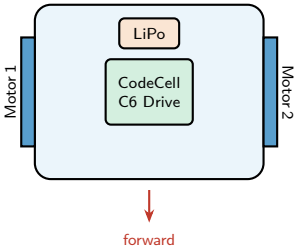
1. Project Overview
2. Hardware
3. Software Architecture
4. Movement Control
5. Wireless Communication
6. Control Application
7. Safety & Robustness
8. Summary & Future Work

Project Goals

Build a **compact, wirelessly controlled** differential-drive robot using the CodeCell C6 Drive as the sole electronic module.

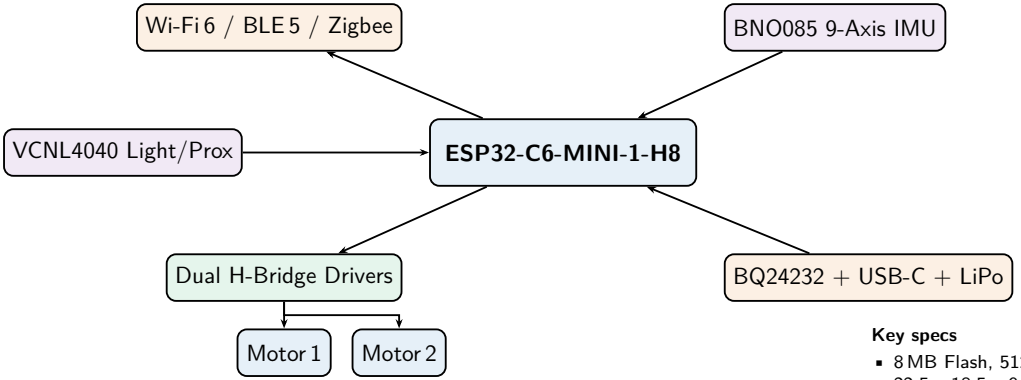
Key objectives:

- Heading-stabilised motion via onboard IMU
- Two operating modes: real-time joystick & autonomous point-to-point
- Browser-based control app (no install needed)
- 3D-printed chassis – reproducible build



Top-down schematic view

CodeCell C6 Drive – Board Overview



Key specs

- 8 MB Flash, 512 KB SRAM
- $22.5 \times 18.5 \times 9.4$ mm
- 3.4 g (board only)
- DRV8837 H-Bridge per channel
- Sensor-fusion on BNO085

Mechanical Design

Differential-drive layout

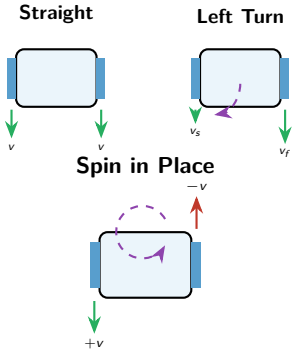
- Left motor → left wheel
- Right motor → right wheel
- Board and LiPo mounted centrally

Chassis

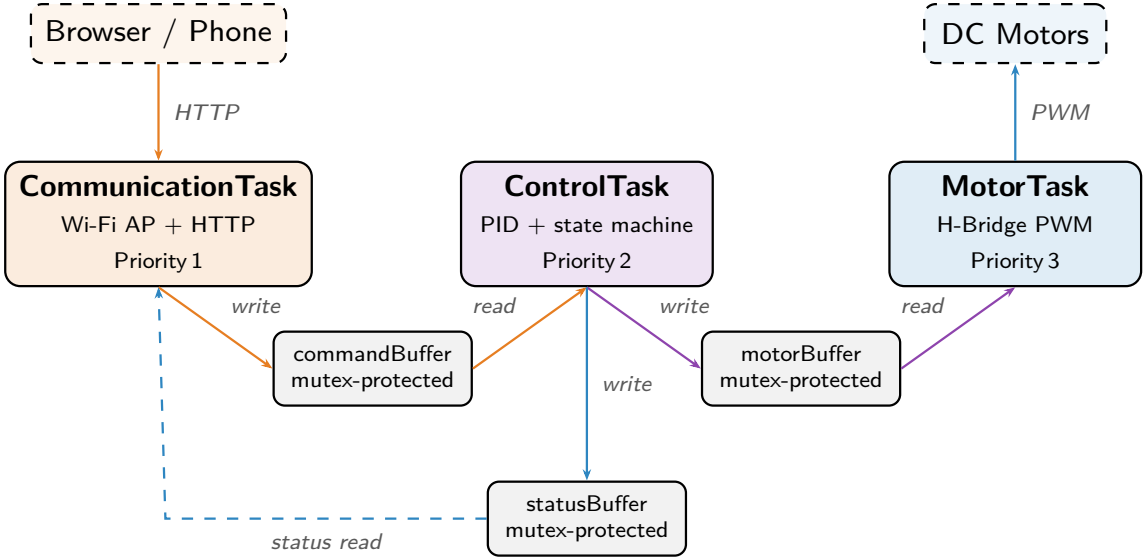
- 3D-printed (FDM), single-piece shell
- Designed for press-fit motor mounting

Turning principle

Equal speed ⇒ straight line
Speed difference ⇒ curved path
Opposite speeds ⇒ rotation in place



FreeRTOS Task Architecture



Differential-Drive Motor Mixing

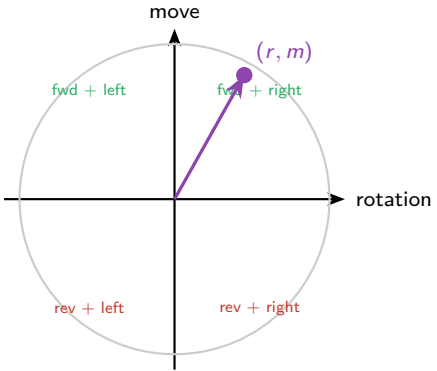
Joystick inputs in $[-100, 100]$:

- `move`: forward/backward
- `rotation`: turning

Mixing equations

$$m_1 = \text{clip}(\texttt{move} + \texttt{rot})$$
$$m_2 = \text{clip}(\texttt{move} - \texttt{rot})$$

where $\text{clip}(x) \in [-100, 100]$. Sign sets direction; magnitude sets PWM duty cycle.



Example

move $= 70$, rot $= 30$

$m_1 = 70 + 30 = 100$

$m_2 = 70 - 30 = 40$

$\Rightarrow$ gentle right curve

Heading-Stabilised Control – PID on Yaw

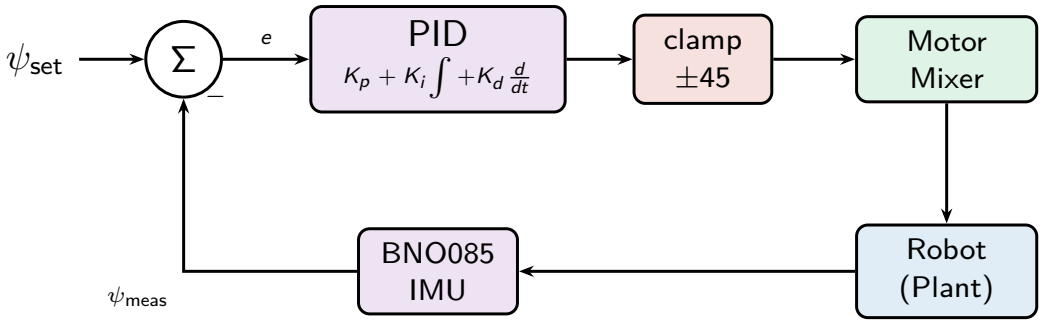
The BNO085 provides fused **yaw** (heading) at each control cycle ($\sim$50 Hz). A discrete PID controller minimises heading error and feeds its output as the `rotation` input to the motor mixer.

Tuning parameters

Parameter	Value
K_p	0.30
K_i	0.02
K_d	0.02
Integral limit	± 120
Max correction	± 45

Error is wrapped to $(-180^\circ, 180^\circ]$ so the controller always uses the shortest rotation path.

PID Yaw Feedback Loop



Control Modes – Direct vs. Target

Both modes are selected by the REST API command sent to the robot.

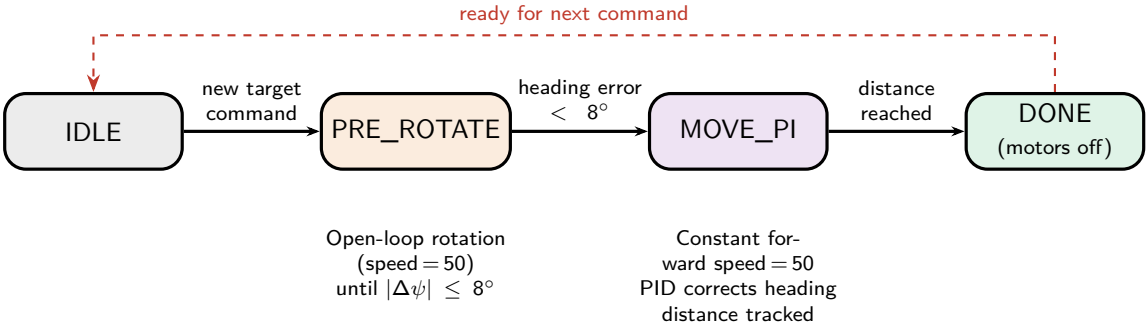
Direct Mode (Joystick)

- Joystick sends (`move`, `rotation`) at 10 Hz via HTTP
- `rotation` $\rightarrow$ yaw rate (degrees/s scaling)
- PID maintains heading setpoint
- Stick release $\rightarrow$ auto-centre, sends $(0, 0) \rightarrow$ stop

Target Mode (Autonomous)

- User specifies distance (m) and angle ($^\circ$)
- Phase 1: rotate to target heading
- Phase 2: drive straight with PID yaw correction
- Distance by timer *or* IMU

Target Mode – Two-Phase Execution



Distance estimation: time-based calibration or IMU double-integration.

Start Ramp & Motor Deadband Compensation

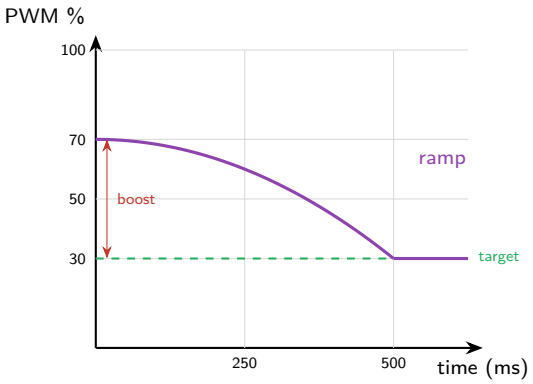
Small DC motors need extra torque to overcome **static friction** at startup.

Start-ramp algorithm

$$v_{\text{out}}(t) = |v_{\text{target}}| + (v_{\text{boost}} - |v_{\text{target}}|) \cdot (1 - t^2)$$

where $v_{\text{boost}} = 70\%$, ramp duration $T = 500\,\text{ms}$, and $t = \text{elapsed}/T$.

The output is clamped to 100 and returns to the requested speed after 500 ms.



Wi-Fi Access Point & REST API

The robot creates its own **Wi-Fi hotspot**; the browser connects directly.

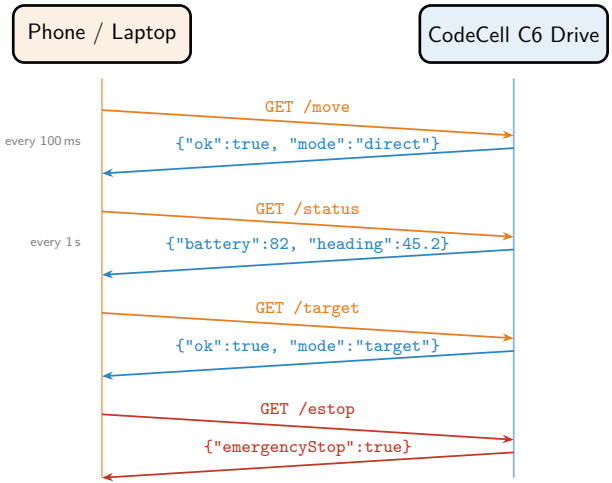
Setting	Value
SSID	`CodeCell-C6-Drive`
Password	`12345678`
IP	`192.168.4.1`
Protocol	HTTP (port 80)

REST commands use HTTP `GET` requests and JSON responses.

API Endpoints

Endpoint	Purpose
`/move`	Joystick speed command
`/target`	Distance + angle command
`/status`	Battery, heading, motor state
`/estop`	Emergency stop
`/clear_estop`	Clear e-stop

Communication Flow



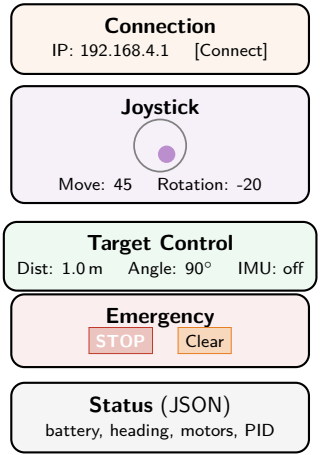
Web-Based Control Interface

Single-page HTML/JS app

- Runs in any modern browser
- Touch-friendly virtual joystick
- Mouse and touch pointer events

Features

- Joystick sends `/move` at 10 Hz
- Target panel: distance, angle, IMU toggle
- Emergency stop / clear buttons
- Live status refreshed every second



Safety Mechanisms

Emergency Stop

- One-click E-Stop via `/estop`
- All tasks check one shared flag
- Cleared only by explicit `/clear_estop`

Command Timeout

- Every command is timestamped
- No update within **300 ms** $\rightarrow$ stop
- Handles Wi-Fi or browser failure

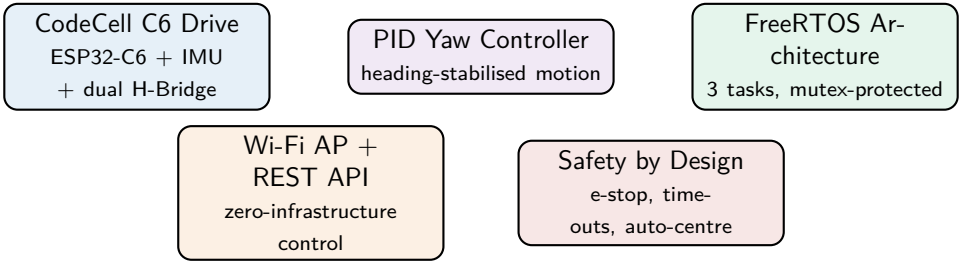
Thread Safety

- Three FreeRTOS mutexes guard shared buffers
- No direct cross-task memory access
- Sequence counter detects new commands

Auto-Centre Joystick

- On `pointerup`, joystick snaps to $(0, 0)$
- Sends stop command immediately
- Prevents unintended continued motion

Summary



A self-contained differential-drive robot built from one compact board, controlled from a browser, with real-time heading stabilisation from the onboard 9-axis IMU.

Future Work

- **Obstacle avoidance** – reactive stop / turn behaviours
- **Odometry improvements** – wheel encoders or visual odometry
- **Multi-waypoint navigation** – sequence of target commands
- **BLE / Zigbee control** – lower-power communication options
- **Smartphone app** – richer UI and persistent settings

Thank you

Questions?

Source code & 3D models available in the project repository